

System Specifications: CogniSync

CogniSync is a specialized web-based Generative AI intervention tool designed to address the mental health challenges associated with technology addiction among Filipino students. The system acts as a digital wellness coach, providing localized support to help users regain control over their digital habits.

1. Functional Specifications

A. Strict Operational Guardrails

- **Geographic Limitation:** The system is strictly limited to the Philippines. It utilizes localized data regarding the Philippine educational system, holiday schedules, and local mental health resources.
- **Domain Limitation:** The AI is strictly limited to Mental Digital Addiction and will refuse to answer technical hardware questions, general academic tutoring, or non-related medical queries.
- **Tone & Language:** The system supports Taglish (Tagalog-English code-switching) to ensure the intervention feels relatable to Filipino students.
- **Tone & Language:** The system is capable of processing and understanding user inputs in Tagalog, Taglish, or English to ensure maximum accessibility. However, the AI formulate all of its responses strictly in English to maintain a clear and professional intervention tone.

B. Core AI Capabilities

- **Symptom Identification:** The AI recognizes and explains symptoms such as Phantom Vibration Syndrome, Nomophobia, and Digital Burnout.
- **Standardized Mitigation:** The system generates general cognitive strategies, such as CBT-based prompts, rather than personalized habit tracking.
- **Exit Strategies:** Provides actionable tips to move away from devices by suggesting popular Philippine "analog" hobbies like local sports or traditional board games.

C. Mitigation & Intervention Features

- **Standardized Action Plans:** The AI provides general structured plans based on common student schedules.
- **CBT-Lite Prompts:** Uses basic Cognitive Behavioral Therapy (CBT) techniques to help students challenge addictive thought patterns.

2. Technical Specifications

Component	Specification
Core Model	Large Language Model (Llama-3.1-8B-Instant) hosted via the Groq API, controlled via strict System Prompt Engineering. With a "Digital Wellness" persona.
Intervention Logic	Retrieval-Augmented Generation (RAG) pipeline powered by LangChain, HuggingFace Embeddings (all-MiniLM-L6-v2), and a FAISS local vector database of Philippine mental health research (2020–2026).
Platform	Web-based (React.js) optimized for mobile browser use.

Frontend	React.js - High-performance, responsive chat interface.
Backend API	Python FastAPI - Handles asynchronous API requests, RAG retrieval, and AI model communication.
AI Engine	Llama-3.1-8B-Instant - Fine-tuned to enforce Philippine and Mental Health boundaries.
Deployment	Vercel - Utilized for frontend hosting and serverless functions.
Design & Prototyping	Figma - Utilized for UI/UX design, wireframing, and high-fidelity prototyping.
Data & Security	Supabase - Integrated for future secure edge-function routing, rate-limiting, and storing anonymous, non-personalized interaction analytics.

3. Scope & Delimitations

What CogniSync IS:

- **Mental Focused:** It addresses the mental urge to use tech (dependency, anxiety, social validation) rather than physical hardware issues.
- **Student-Centric:** Content is tailored for Filipino high school and college students.
- **Preventative:** Aimed at helping "at-risk" users before they reach clinical levels of addiction.

What CogniSync IS NOT:

- **Medical Diagnosis:** It does not provide medical prescriptions or formal psychiatric diagnoses.
- **Personalized:** The system does not track, store, or adapt to individual user behavior, maintaining a standardized intervention approach.
- **Content Filter:** It does not block apps or websites; it provides the *mental tools* for the student to do so themselves.
- **General Purpose AI:** It will refuse to answer questions about non-related topics (e.g., math homework, global news, or physical ailments).

4. Example System Output (User Interaction)

Student: "Hindi ako makatulog kasi feeling ko I'm missing out sa mga chika sa GC namin."

CogniSync: "It sounds like you're experiencing FOMO, which is common among students using messaging apps late at night. Did you know that 77% of teens feel this same anxiety? (Kemp, 2024). Let's try a 'Soft Disconnect': Set your phone to 'Do Not Disturb' for just 20 minutes. The drama will still be there tomorrow, but your brain needs the rest to handle schoolwork. Want to try a 2-minute breathing exercise instead?"